Influence of Stimulus Color on Steady State Visual Evoked Potentials

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Abstract Due to the low training time and high time resolution, steady state visual evoked potential (SSVEP)-based brain computer interfaces (BCIs) have been largely studied in recent years. The stimulus properties such as frequency, color and shape can greatly affect the performance, comfort and safety of the brain computer interfaces. Despite this fact, stimulation properties have received fairly little attention. This study aims to investigate the influence of stimulus color on SSVEPs. We tested 10 colors and did evaluation by using three different methods: the power amplitude, multivariate synchronization index, and canonical correlation analysis. The results showed that violet color had the least influence to SSVEPs while red color tended to have stronger impact.

Keywords SSVEP · BCI · EEG · Stimulus properties

1 Introduction

Brain computer interface (BCI) is a technology that allows communication between machines and the human brain. This technology allows people with neurodegenerative diseases or spinal cord injury to control robots such as wheelchairs and prosthetic devices by brain signals.

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Studies about BCIs have shown promising results in recent years. Patients with complete hand paralysis were able to close and open a virtual hand by focusing their attention [1]. Performance of a mental spelling system, was enhanced by implementing a dual eye tracking/BCI system [2]. A wheelchair prototype was able to be controlled with high accuracies by brain signals [3]. Although the performance of BCIs is enhancing, more effort is needed to assure the stability and accuracy of BCIs so that they are applicable in real life. In addition, the performance of BCIs also rely on the physical conditions and physiological traits such as, concentration or surroundings between others [4]. One type of response that is relatively easy to detect is the Steady State Visual Evoked Potential (SSVEP) in electroencephalogram (EEG). SSVEP is a response that can be detected at visual field of cerebral cortex when a flickering light stimulates the retina. This response occurs at the same frequency as the flickering stimulus and, with lower intensity, in the harmonic and sub-harmonic frequencies [5]. SSVEP-based BCIs provide promising result because of their short training time, easy detection and accessibility for most people [6].

Although SSVEP-based BCIs have been studied in recent years, most studies focus on the signal processing techniques to improve the performance of BCIs. Despite the fact that parameters of stimulus properties such as shape, color and size have significant effects on the performance and safety of SSVEP-based BCIs [4], stimulus properties are given fairly little attention. It is important to study which parameters provide the most efficient BCI systems to optimize the performance of SSVEP-based BCIs. The main objective of this study is to study the influence of stimulus colors to SSVEP.

Few studies focus their attention on the effect of the color to the SSVEP response. Most studies were done only with a few colors (e.g. yellow, green, red, white, violet) [3, 4, 7] which we consider insufficient to conclude which color is more effective to SSVEPs. In addition, most studies about performance of stimuli colors on SSVEPs were done by displaying a multiple flickers simultaneously [3, 7]. Stimuli with larger size and closer to one another has detrimental effect on SSVEP performance of the target stimulus [4]. The studies of [8] showed that inter-stimulus distance is significantly different with SSVEP performance. This might due to interference of neighboring flickers on the same visual field and this may cause users to lose their attention. To tackle these problems, we investigated the influence of wavelength of colors and their respective color luminance to SSVEPs with nine colors from the visible spectrum (380–735 nm) and white color, which in total ten colors. Our study focuses on the sheer SSVEP response to each color, therefore, we displayed only one visual stimulus during the stimulation of each color.

2 Materials and Methods

2.1 *Participants*

Fifteen healthy subjects (nine males, six females) participated in our experiment. All subjects have normal or corrected-to-normal vision and did not suffer from epilepsy. The mean age is 23.9 years old in the range of 19–25 years old. Permission of the ethics committees of Chiba University Graduate School and Faculty of Engineering was obtained. All subjects participated voluntarily with informed consent and were informed that they could leave the experiment if they experience discomfort or epilepsy during the experiment.

2.2 *Stimulation*

We used a 10 Hz stimulation frequency. The flickering stimulus was generated using a laptop through a control program written in C++ running on Ubuntu 14.04.3 LTS. A 22 in. monitor screen (LG W2242TQ-BF), a 60 Hz refresh rate and a resolution of 1680 × 1050 pixels adjusted to its brightest state (300 cd/m^2) was used to present the stimulus. Colors were chosen within the range of visible spectrum from 390–735 nm as shown in Table 1. Within the range, we intended to select nine colors which have equal difference in wavelength from one another, which is 40 nm. However, some colors appear to have no difference due to the limitations of our monitor display. In that case, we selected colors which have difference within 25 nm from 40 nm and appear differently from the previous color. Note that the colors we obtained are of approximation. Luminance of the color was

Table 1 Color information

	Color	Wavelength (nm)	RGB	Luminance
1		390	(101,0,121)	30.21
2		430	(43,0,255)	27.55
3		470	(0,153,255)	127.84
4		510	(0,255,0)	182.38
5		550	(146,255,0)	213.42
6		600	(255,177,0)	180.80
7		630	(255,59,0)	96.41
8		670	(255,0,0)	54.21
9		735	(177,0,0)	37.63
10		-	(255,255,255)	255.00

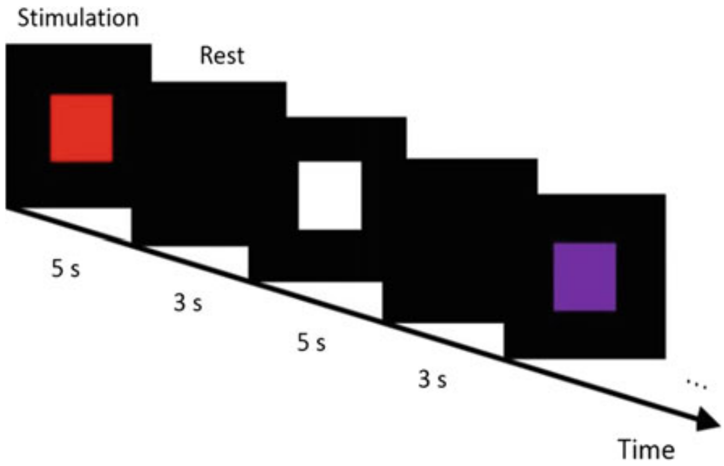

Fig. 1 Stimulation sequences in experiment

calculated respectively according to Recommendation ITU-R BT.709-6 [9], under consideration that our monitor display uses a sRGB color space.

$$\text{Luminance} = 0.2126\text{R} + 0.7152\text{G} + 0.0722\text{B} \quad (1)$$

The experiment was carried out in a dark room, with the subjects seated approximately 70 cm from the center of the monitor screen. Subjects were asked to ocular or muscular movements during visual stimulation. A stimulus with a size of 500 × 500 pixels was presented at the center of monitor display during stimulation. Every color was presented for 5 s with a 3 s resting time in between without any stimulus (Fig. 1). The process was repeated for the 10 colors in random order. Stimulating the 10 colors take 80 s. This process was repeated 10 times for every subject.

2.3 EEG Measurement

We acquired EEG by Biosemi ActiveTwo system (Biosemi, Netherlands) with 16 active channels. The location of the electrodes was chosen to primarily cover the parietal and the occipital area. The electrode positions were: T7, T8, C3, C4, Cz, P1, P2, P3, P4, Pz, PO3, PO4, POz, O1, O2 and Oz according to the international 10–20 system. The signal acquisition frequency of the system was 1024 Hz.

2.4 Signal Processing

Previous study found that detection of SSVEP is better by using two bipolar channels (e.g. PO3-O1) than by using monopolar channels (e.g. O1, O2 etc.) [10]. Therefore, although 16 EEG channels were recorded we used only channels POz and Oz (POz-Oz) for the signal processing. They were filtered with by a 4th order

Butterworth high pass filter with cutoff frequency of 1 Hz. Then they were divided into 1 s segment with 87.5 % overlap. For every segment a detrending and a hamming window was applied. Next, we evaluated SSVEP with SSVEP power amplitude, multivariate synchronization index (MSI), and canonical correlation analysis (CCA). All the analysis were performed offline.

SSVEP Power Amplitude

For every segment of the pre-processed data the corresponding FFT was calculated. The highest power amplitude for the 10 Hz frequency, during stimulation of every color for every repetition was selected. In total we had 10 values for each one of the 10 colors for each subject, i.e. 100 values/subject.

MSI

MSI is a novel classification method for frequency recognition in SSVEP. MSI is used to estimate the synchronization between the EEG signals and the reference signal [11], which in our case, generated from the sine and cosine components of the 10 Hz stimulus. Assume that the EEG signals are denoted by a matrix X of size $N \times M$, and the reference signal by a matrix Y of size $2N_h \times M$. N is the number of channels, M is the number of samples and N_h is the number of harmonics for the sine and cosine components. In our case, we used only channel POz and Oz, therefore $N = 2$. We also divided the data into 1 s segments, corresponding to our sampling frequency which is 1024 Hz, $M = 1024$. We used only one harmonic frequency therefore $N_h = 1$. Next, a correlation matrix is calculated as

$$\mathrm{c} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \tag{2}$$

where

$$C_{11} = \frac{1}{M} XX^T \tag{3}$$

$$C_{22} = \frac{1}{M} YY^T \tag{4}$$

$$C_{12} = {C_{21}}^T = \frac{1}{M} XY^T \tag{5}$$

To remove the internal correlation structure of X and Y which is irrelevant to the stimulus frequency, a linear transformation is implemented [12].

$$U = \begin{bmatrix} C_{11}^{-\frac{1}{2}} & 0 \\ 0 & C_{22}^{-\frac{1}{2}} \end{bmatrix} \tag{6}$$

Then, the transformed correlation matrix R is calculated.

$$R = \mathrm{UC}U^T = \begin{bmatrix} I_{N\times N} & C_{11}^{-\frac{1}{2}} C_{12} C_{22}^{-\frac{1}{2}} \\ C_{22}^{-\frac{1}{2}} C_{21} C_{11}^{-\frac{1}{2}} & I_{2N_h \times 2N_h} \end{bmatrix} \tag{7}$$

where $I_{N\times N}$ is the identity matrix of dimension N, and $I_{2N_h \times 2N_h}$ is the identity matrix of dimension $2N_h$. Then, normalized eigenvalues are calculated from the eigenvalues, λ_1, λ_2, …, λ_P of matrix R.

$$\lambda_i^{'} = \frac{\lambda_i}{\sum_{i=1}^{P} \lambda_i} = \frac{\lambda_i}{tr(R)} \tag{8}$$

where $P = N + 2N_h$. Then the synchronization index between the EEG signals and the reference signal can be calculated.

$$\text{Synchronization index} = 1 + \frac{\sum_{i=1}^{P} \lambda_i^{'} \log(\lambda_i^{'})}{\log(P)} \tag{9}$$

Consider that there is only one stimulus frequency, which is 10 Hz in our experiment, the reference signal is calculated as follows:

$$Y_i = \begin{bmatrix} \sin(2\pi \cdot f \cdot t_i) \\ \cos(2\pi \cdot f \cdot t_i) \\ \vdots \\ \sin(2\pi \cdot N_h \cdot f \cdot t_i) \\ \cos(2\pi \cdot N_h \cdot f \cdot t_i) \end{bmatrix} = \begin{bmatrix} \sin(2\pi \cdot 10 \cdot t_i) \\ \cos(2\pi \cdot 10 \cdot t_i) \end{bmatrix},\ t_i = \frac{i}{F_s} \tag{10}$$

$$Y = [Y_1 Y_2 \ldots Y_i],\ i = 1, 2, \ldots, M \tag{11}$$

where F_s is the sampling frequency, which is 1024 Hz. Note that $N_h = 1$ as aforementioned. Synchronization index is used as an indication of how well the EEG signals synchronized with the stimulus and we named it as MSI value in the following sections. In each repetition, maximum MSI value was selected for every color.

CCA

CCA is a method for SSVEP frequency recognition, which is a multivariable method to compute the correlation between two sets of data. It is widely used in studies to extract frequency features for multichannel SSVEP-based BCI [13–15]. First, assume that we have x_1, x_2,…, x_i which are signals from i number of EEG channels, and y_1, y_2,…, y_j which are signals from j number of reference signals as calculated in Eqs. (10) and (11). Two data sets, X and Y with linear combinations $w_{x_1}x_1$,…, $w_{x_i}x_i$ and $w_{y_1}y_1$,…, $w_{y_j}y_j$ were calculated respectively as follows:

$$X = w_{x_1}x_1 + \ldots + w_{x_i}x_i = {w_x}^T x \quad (12)$$

$$Y = w_{y_1}y_1 + \ldots + w_{y_j}y_j = {w_y}^T y \quad (13)$$

where $w_{x_1},\ldots, w_{x_i}$ and $w_{y_1},\ldots, w_{y_j}$ are the linear combination coefficients. The CCA finds the linear combination coefficients which maximize the correlation between X and Y. The correlation coefficient is then calculated as follows:

$$\rho = \frac{w_x^T x y^T w_y}{\sqrt{w_x^T x x^T w_x w_y^T y y^T w_y}} \quad (14)$$

CCA is discussed more in detail by [13, 14]. Similar as in SSVEP power amplitude and MSI, the maximum value of CCA coefficient for every color was selected in every repetition.

Statistics

After extracting the maximum value of each color for every repetition by all three methods, we examined correlation of SSVEP and color luminance, and correlation of SSVEP and colors (wavelength). To examine the correlation of SSVEP and color luminance, first we calculated the mean values of all the repetitions for every color and subject. As we have fifteen subjects, there are 15 mean values for each method. Finally, the mean of these 15 values was calculated for every color. Then, we computed the Pearson's correlation coefficient to examine the relationship between SSVEP and luminance. The significance threshold for the analysis was assigned as $p < 0.05$. About correlation of SSVEP and colors (wavelength), we compared the influence of colors on SSVEP by Friedman test followed by Tukey-Kramer post hoc test. The significance threshold for Friedman test was assigned as $x^2 < 0.05$.

3 Results

3.1 Correlation of SSVEP and Luminance

Figure 2 shows correlation between SSVEP and color luminance by (a) SSVEP power amplitude, (b) MSI, and (c) CCA. The colors of the dots represent the stimulation colors. By SSVEP power amplitude, the correlation coefficient, r = 0.2921, $p = 0.4128$. By MSI, r = 0.5698, $p = 0.0855$. By CCA, r = 0.5577, $p = 0.0939$. From the results, p by all methods are higher than the significance level of 0.05, showing that SSVEP is not significantly related to luminance.

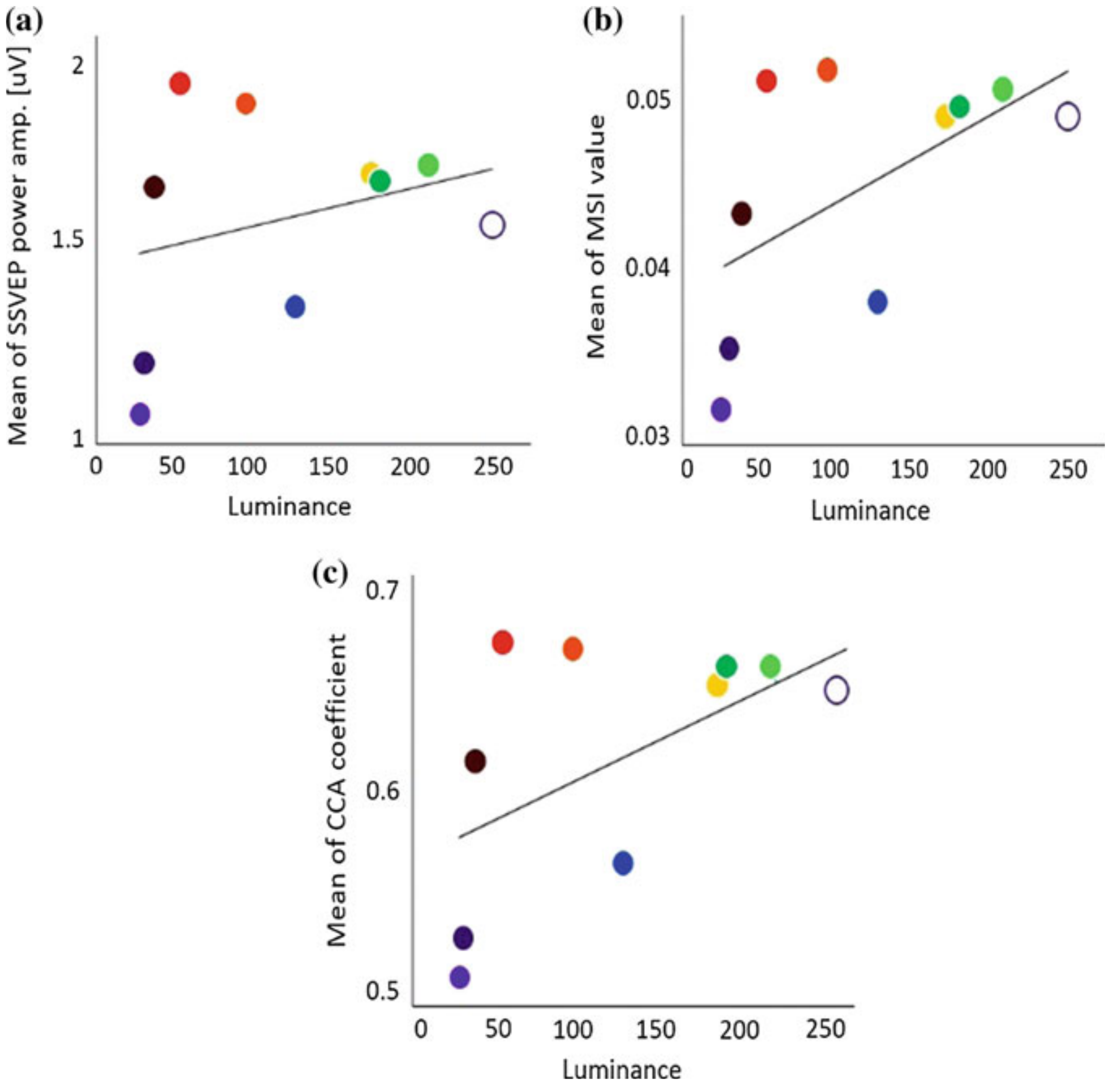

Fig. 2 Correlation of SSVEP and luminance, **a** SSVEP power amplitude, **b** MSI, **c** CCA

3.2 Correlation of SSVEP and Colors (Wavelength)

Figure 3 shows the results of such comparison by (a) SSVEP power amplitude, (b) MSI, (c) CCA. The colors of the bars represent the stimulation colors. With all the methods we used, color 2 is significantly different with other 5 colors (color 4, 5, 6, 7 and 8). The means of color 1 and color 3 are also less than the other colors although they are not statistically significant different except for SSVEP power amplitude that color 1 is significantly different with color 4, 5, 6, 7 and 8. Although all the results appear to be similar, result with SSVEP power amplitude is slightly different with the other two methods. With SSVEP power amplitude, color 7 (orange) and color 8 (red) appear to have higher mean compared to other colors.

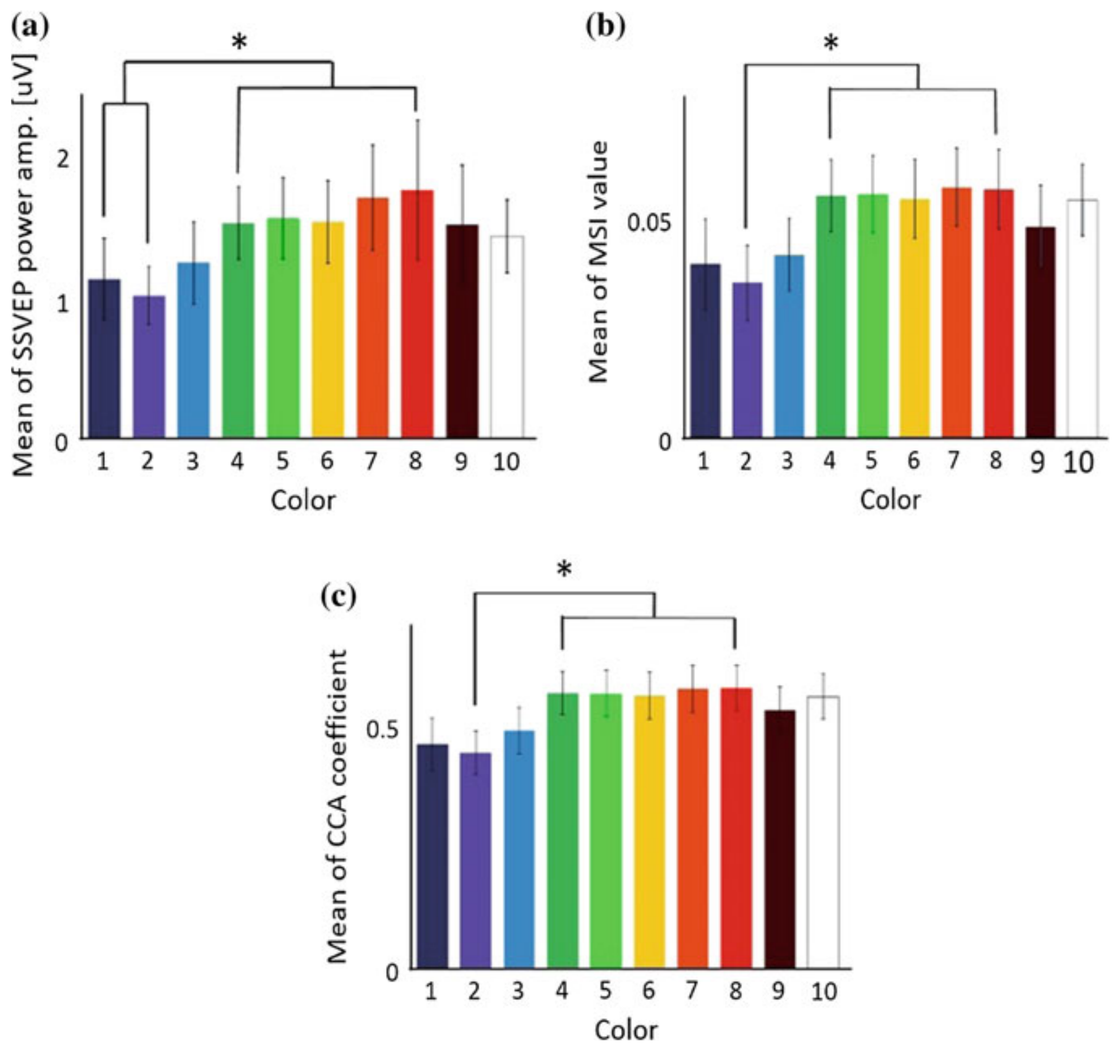

Fig. 3 Error bars with means and 95 % confidence intervals (* χ^2 < 0.05 Friedman test followed by a Tukey-Kramer post hoc test), **a** SSVEP power amplitude, **b** MSI, **c** CCA

4 Discussion

We investigated the relation between SSVEP and color and found that violet color has the worst SSVEP response. Our results contradicted with [3], who showed that violet color has higher accuracy than that with green, red and blue color They displayed four stimuli simultaneously with four different frequencies and tested the accuracy to control a wheelchair prototype, while our experiment simply tested SSVEP response by displaying only one stimulus each time. The difference on the results may arise from the use of multiple colors at the same time, as well as the use of different frequencies.

Colors with longer wavelength such as color 7 (orange color) and color 8 (red color) appear to have higher mean compared to other colors, although the differences are not significant. Since looking at a flickering stimulus and generating SSVEPs requires high visual attention, it is recommended to use colors that may guide attention. The studies of [16] are coherent with our results that warm colors such as red and orange are more likely to guide attention. It is no doubt that red color grabs attention and stimulates visual sense considering its usage on symbolic sign like “dangerous” and “attention” to draw people attention. In this study the results also show that red color might have a great effect on SSVEPs, which are considered as a type of attention-related brain signals. The result is supported by the studies of [17] about response of red color in human eyes, by showing that red color had the prominent amplitude peak compared to blue and yellow color. Red color also provides the highest rates of accuracy for detection of SSVEP as showed by [7]. Our similar results indicate that by displaying only one stimulus each time could be helpful in preventing interference by neighboring flickers.

In our study, it can be concluded that colors with shorter wavelength like violet and blue have relatively worse SSVEP response compared to other colors. On the other hand, colors with longer wavelength like orange and red tend to have better SSVEP response. However, our study investigated the sheer SSVEP response to colors and only offline experiment was done. As SSVEP-based BCIs are involved in real life application, online experiment is necessary to give further insights about our results. As our experiment displayed only one stimulus each stimulation, displaying multiple stimuli with different colors simultaneously might be the future work to further investigate if it supports our current results. Besides, different stimulation frequencies might be tested to study the correlation between frequency and color, and their effect on SSVEPs. In addition, questionnaire will be carried out at the end of the experiment to study the comfort level of each color in each subject.

The future study is expected to provide us more understanding and convincing result to the influence of colors to SSVEPs.

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